

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Digital Campus Library Reservation Gateway — UC-04 Pay Overdue Fine

Student	Dharshini V
SRN	PES1UG24AM428
Use Case	UC-04 — Pay Overdue Fine
Architecture	Layered Architecture

1. Scenario Review

UC-04 allows a Student Member with an outstanding overdue fine to view itemized fines, choose a payment method, submit payment through an external Payment Gateway, receive a receipt, and have the account balance reduced to zero after successful authorization. If payment is declined, the fine remains outstanding and the student can retry.

2. Architectural Style Analysis

Layered Architecture provides clear separation between presentation, business logic, and data/integration concerns. Microservices could provide independent scaling and fault isolation, but would introduce additional operational complexity and network/data-consistency overhead. Client-Server would provide centralized control, but a centralized server can become a scalability and availability bottleneck.

3. Architecture Selection

We chose Layered Architecture for the Digital Campus Library Reservation Gateway, specifically for the UC-04 Pay Overdue Fine workflow.

Reason 1: The Student Account UI, Fine Management Service, Payment Service, account database, and receipt/notification functions have distinct responsibilities. This makes the payment workflow easier to test, maintain, and change without modifying unrelated parts.

Reason 2: The Payment Service can isolate communication with the external Payment Gateway from the student-facing interface and library account data. This is appropriate because UC-04 explicitly requires a Payment Gateway interaction and a successful payment must update the fine balance and generate a receipt.

Security Advantage: The architecture provides controlled boundaries around payment processing and account data. The UI does not directly modify the database or communicate directly with the external Payment Gateway; requests pass through the Fine Management and Payment Service components.

Performance Benefit: The Fine Management Service can coordinate local account/fine queries and payment workflow without unnecessary distributed-service hops. The account database and receipt/notification functions can also be optimized independently while keeping the student-facing workflow responsive.

4. Components and Interfaces

Five core components are modelled: Student Account UI, Fine Management Service, Payment Service, Library Account Database, and Receipt & Notification Service. The external Payment Gateway is additionally shown because it is explicitly a supporting actor in UC-04. Key interfaces include Fine Inquiry / Pay Fine, Payment Request, Fine & Account Query, Payment API, Authorization Status, Receipt Request, and Receipt / Email.

5. UML Component Diagram

Digital Campus Library Reservation Gateway — UC-04 Pay Overdue Fine

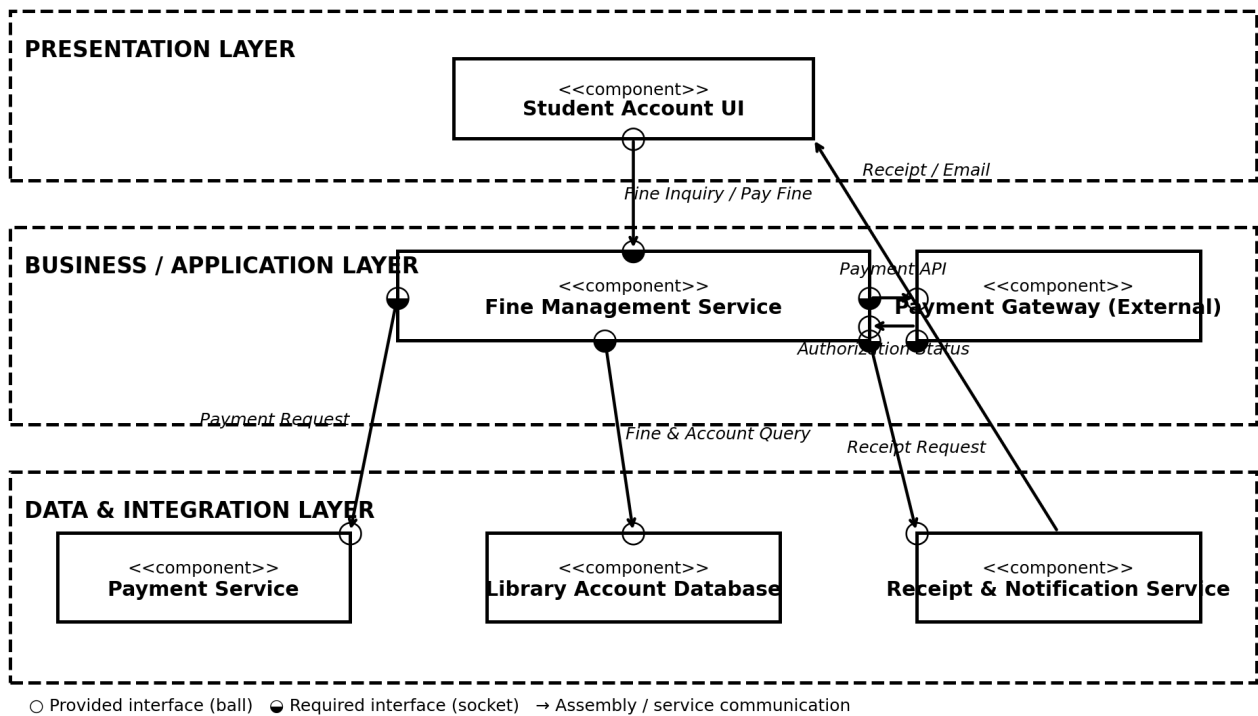


Figure 1. UC-04 Pay Overdue Fine — Layered Architecture Component Diagram

The diagram uses component notation and distinguishes provided/required interface points in the legend. The external Payment Gateway is shown outside the core layered components.